Lab Assignment 2
Task 2b: Requirements and Planning

SmartBrew AI

User Stories

- As a Customer, I want to look at the digital menu so that I can see the coffee choices and prices.
- As a Customer, I want to use sliders to change the amount of sugar and milk so that the drink tastes the way I like.
- As a Customer, I want to pay with a contactless card or phone so that paying is fast and easy.
- As a Customer, I want the AI to suggest a drink based on the weather and time so that I can try something new. **(Custom Feature)**
- As a Customer, I want to see a progress bar so that I know when my coffee is ready.
- As a System Admin, I want to edit the menu so that I can add new drinks and keep it updated.
- As a System Admin, I want to get an alert when ingredients are low (below 15%) so that I can refill the machine in time.
- As a System Admin, I want to see daily sales reports so that I know which drinks sell best.
- As a Maintenance Technician, I want to run hardware tests on pumps and heaters so that I can find broken parts easily.
- As a Maintenance Technician, I want to save error logs to a USB drive so that I can check them on my computer.

System Requirements

ID	Description	Inputs	Outputs	Precondition	Postcondition
REQ-01	Display digital menu and customization options	Touch interaction on the main screen	Drink selection UI and milk/sugar sliders	System is powered on	Menu is displayed and drink is selected
REQ-02	Process contactless payment	NFC card/phone tap	Payment success or failure notification	Drink is selected and total price is calculated	Transaction is approved and brewing begins
REQ-03	AI drink suggestion	Time of day and weather API data	Recommended drink name	Internet connection is active	Suggestion is displayed to the customer

REQ-04	Monitor inventory levels	Ingredient sensor data	Low-level alert notification	Machine has active physical sensors	Admin is notified if any level drops below 15%
REQ-05	Admin menu management	Admin login credentials and new menu item data	Updated digital menu	Admin is logged into the system	New drinks are visible on the customer menu
REQ-06	Hardware maintenance mode	Maintenance PIN and component selection	Test execution results and error logs	System is locked in service mode	Hardware is tested and logs are saved to USB

Initial Product Backlog

Note: The Importance scale ranges from 1 to 100 (100 being the highest priority). Effort is estimated in Story Points following the standard Fibonacci sequence (1, 2, 3, 5, 8).

ID	Name	Imp.	Est.	Description	How to demo	Notes
SB-01	UI Menu & Customization	100	5	Build the touch screen main menu with drink choices and milk/sugar sliders.	Select a "Latte", change the milk slider to 20%, and verify the UI updates the recipe preview.	Core feature
SB-02	NFC Payment Integration	95	5	Add card reader support for contactless payments.	Tap a registered test card, wait for "Approved" message, and verify the order starts.	Needs hardware
SB-03	Weather Data Fetcher	80	3	Implement a background service to fetch current local weather and time.	Disconnect internet to trigger an error, then reconnect and verify weather data is logged.	AI Sub-task 1
SB-04	AI Suggestion Widget	75	5	Build the UI that uses weather data to suggest a specific drink.	Manually set system weather to "30°C", click "Surprise Me", and verify an iced drink is suggested.	AI Sub-task 2
SB-05	Inventory Alerts	90	3	Create a background check that triggers a warning if sensors drop below 15%.	Override the database bean level to 10% and verify an alert appears on the admin panel.	Requires sensor API
SB-06	Admin	85	3	Build a secure	Enter incorrect	Admin

	Authenticat ion			login screen for administrators.	credentials to check error handling, then log in successfully with valid admin details.	Sub-task 1
SB-07	Menu Editor Dashboard	85	5	Build a panel for admins to add or edit drinks.	Log in, add a "Matcha Latte" with price $4, and verify it appears on the customer UI.	Admin Sub-task 2
SB-08	Service Mode Login	70	2	Create a hidden PIN screen for technician access.	Tap the screen corner 3 times, enter PIN, and verify the maintenance interface opens.	Maint. Sub-task 1
SB-09	Hardware Diagnostic UI	70	5	Build manual control buttons to test pumps.	Click "Test Milk Pump", verify the specific pump activates for 2 seconds without brewing coffee.	Maint. Sub-task 2
SB-10	USB Error Log Exporter	60	3	Implement a function to save system error logs to a USB drive.	Insert a USB, click "Export Logs", and verify a CSV file is generated on the drive.	Maint. Sub-task 3